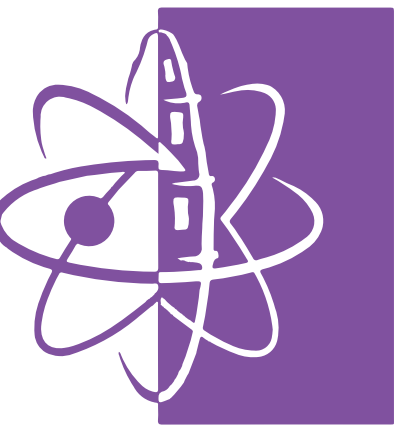




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Medication Tracking Application for Elderly Users

Supervisor
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Abstract

Medication adherence remains a critical challenge in healthcare, particularly among elderly populations who often manage multiple chronic conditions requiring complex medication schedules. Research indicates that approximately 50% of elderly patients do not take their medications as prescribed, leading to adverse health outcomes, increased hospitalizations, and higher healthcare costs estimated at \$100-300 billion annually in the United States alone.

This project develops a user-friendly medication tracking application designed specifically for elderly users and their caregivers. The application is cross-platform deployable as native desktop applications for Windows, macOS, and Linux, as well as a web application accessible through modern browsers.

Features

The system features comprehensive medication schedule management, intelligent reminder notifications, adherence tracking, caregiver monitoring capabilities, and secure local data storage. The interface emphasize accessibility with large fonts, high contrast ratios, simplified navigation, and large touch buttons suitable for elderly users with visual and motor impairments. Key functionalities include flexible medication scheduling supporting various intervals, quick medication entry, drug database integration, drug interaction warnings, medication quantity tracking with refill reminders, health measurement logging (blood pressure, glucose), and professional PDF report generation for healthcare provider consultations. The system supports multiple languages including Arabic and English with proper right-to-left (RTL) text support.

Objectives

Improve Adherence: Reduce missed doses through reliable, persistent reminders and simplified tracking.

Enhance Accessibility: Implement large fonts, high contrast interfaces, and intuitive navigation for seniors.

Empower Caregivers: Enable remote monitoring and instant missed dose alerts for peace of mind.

Privacy First: Prioritize local data storage (SQLite) to ensure sensitive health data remains private.

Cross-Platform: Native support for Windows, macOS, Linux, and Web from a single efficient codebase.

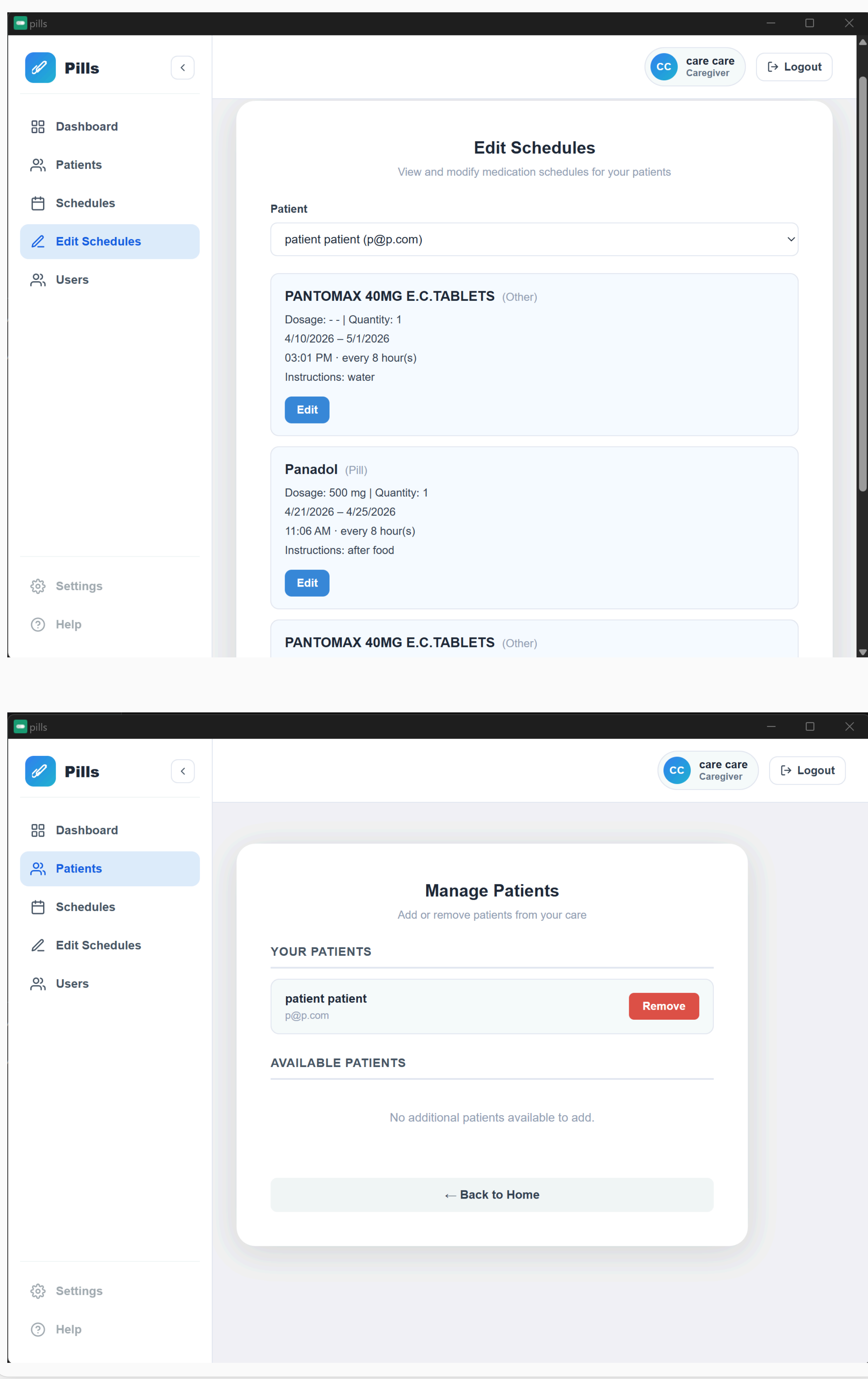
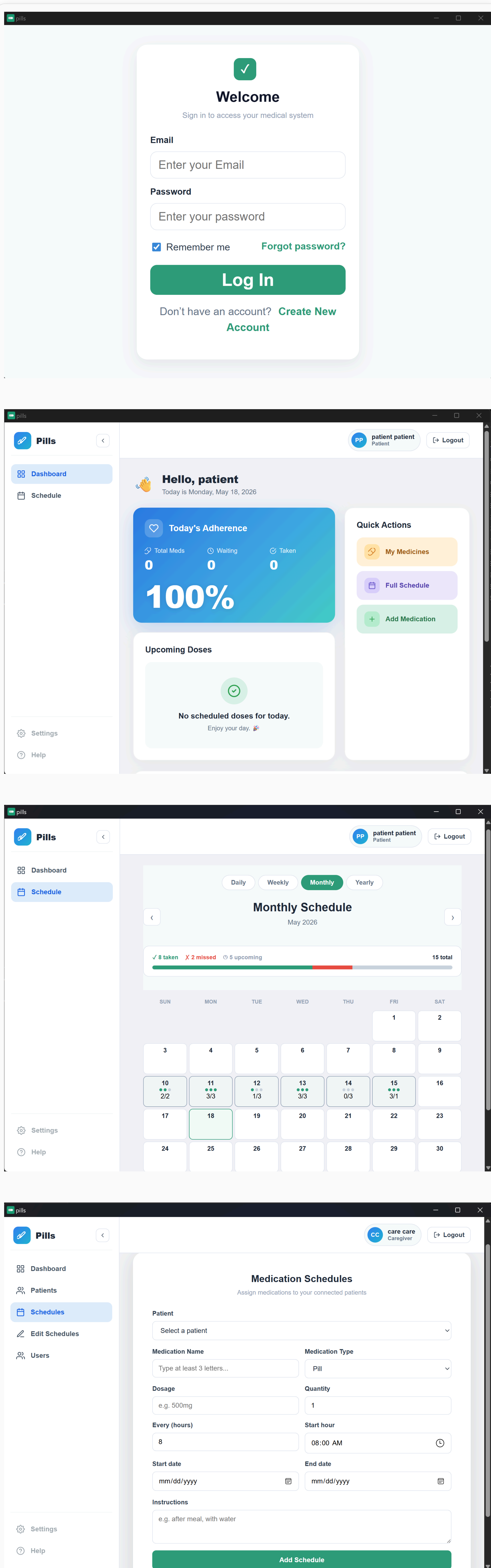
System Architecture

This project involves developing a user-friendly medication tracking application designed specifically for elderly users and their caregivers. The application will be built using Go programming language with the Wails framework to create cross-platform applications deployable as native desktop applications for Windows, macOS, and Linux, as well as a web application accessible through modern browsers. The technical architecture combines a high-performance Go backend for business logic and data management with a Svelte frontend built using modern web technologies (JavaScript, HTML, CSS), enabling dual deployment from a single codebase.

Conclusion

We designed and implemented a solution that bridges the gap between elderly patients and caregivers. By prioritizing privacy, accessibility, and maintaining a free cost model, we addressed some of the shortcomings of current solutions.

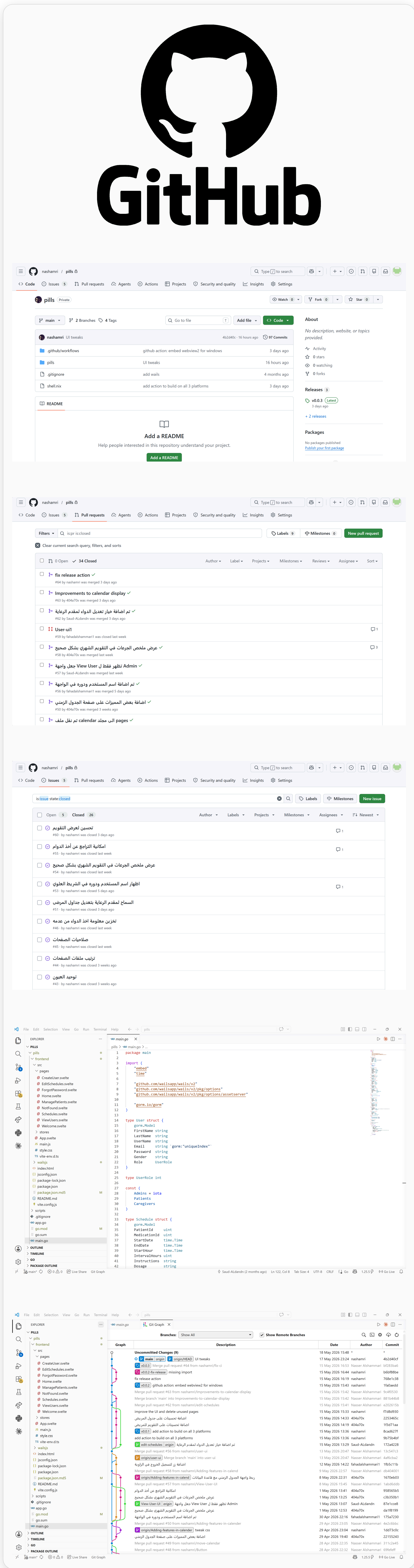
Application Interface



Technologies



Development Workflow



Acknowledgments

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We thank all the project committee members who have sacrificed their time in reviewing this work, and more largely, for the interest which they carried to this project documentation.

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References

- [1] R. L. Cutler, E. Fernandez-Llimos, M. Frommer, C. Bermejo, and V. Garcia-Cardenas, "Economic impact of medication non-adherence by disease groups: a systematic review," *BMJ Open*, vol. 8, no. 1, p. e016982, 2018, doi: 10.1136/bmjopen-2017-016982.
- [2] J. Jin, G. E. Sklar, V. Min Sen Oh, and S. Chuen Li, "Factors affecting therapeutic compliance: A review from the patient's perspective," *Therapeutics and Clinical Risk Management*, vol. 4, no. 1, pp. 269-286, 2008.
- [3] L. Osterberg and T. Blaschke, "Adherence to medication," *New England Journal of Medicine*, vol. 353, no. 5, pp. 487-497, 2005, doi: 10.1056/NEJMr050100.
- [4] A. O. Iuga and M. J. McGuire, "Adherence and health care costs," *Risk Management and Healthcare Policy*, vol. 7, pp. 35-44, 2014, doi: 10.2147/RMHP.S19801.
- [5] New England Healthcare Institute, "Thinking outside the pillbox: A system-wide approach to improving patient medication adherence for chronic disease," Cambridge, MA, Research Brief, 2009.
- [6] M. Viswanathan et al., "Interventions to improve adherence to self-administered medications for chronic diseases in the United States: a systematic review," *Annals of Internal Medicine*, vol. 157, no. 11, pp. 785-795, 2012, doi: 10.7326/0003-483X-157-11-20121230-00538.
- [7] P. J. McDonnell and M. R. Jacobs, "Hospital admissions resulting from preventable adverse drug reactions," *Annals of Pharmacotherapy*, vol. 36, no. 9, pp. 1331-1336, 2002, doi: 10.1345/aph.1A333.
- [8] A. E. Yap, T. Thirumoorthy, and Y. H. Kwan, "Medication adherence in the elderly," *Journal of Clinical Gerontology and Geriatrics*, vol. 7, no. 2, pp. 64-67, 2016, doi: 10.1016/j.jcgg.2015.05.001.
- [9] T. M. Ruppert, P. S. Cooper, D. R. Mehr, J. M. Delgado, and J. M. Dunbar-Jacob, "Medication adherence interventions improve heart failure mortality and readmission rates: Systematic review and meta-analysis of controlled trials," *Journal of the American Heart Association*, vol. 5, no. 6, p. e002606, 2016, doi: 10.1161/JAHA.115.002606.
- [10] N. Mansourn, S. Shakkil, L. Kalisch-Elliott, and C. E. Caughey, "What is polypharmacy? A systematic review of definitions," *BMC Geriatrics*, vol. 17, no. 1, p. 230, 2017, doi: 10.1186/s12877-017-0621-2.
- [11] E. D. Kantor, C. D. Rehm, J. S. Haas, A. T. Chan, and E. J. Giovannucci, "Trends in prescription drug use among adults in the United States from 1999-2012," *JAMA*, vol. 314, no. 17, pp. 1818-1831, 2015, doi: 10.1001/jama.2015.13766.
- [12] A. J. Claxton, J. Cramer, and C. Pierce, "A systematic review of the associations between dose regimens and medication compliance," *Clinical Therapeutics*, vol. 23, no. 8, pp. 1296-1310, 2001, doi: 10.1016/S0149-2918(01)80109-0.
- [13] "Apple," [Online]. Available: <https://www.apple.com/>
- [14] "MacRumors," [Online]. Available: <https://www.macrumors.com/how-to/add-medication-health-app-ios/>
- [15] "medisafe," [Online]. Available: <https://www.medisafe.com/>
- [16] The Go Authors, "The Go Programming Language," [Online]. Available: <https://go.dev/>
- [17] A. A. Donovan and B. W. Kernighan, *The Go Programming Language*. Addison-Wesley Professional, 2015.
- [18] The Go Authors, "Effective Go: Concurrency," [Online]. Available: <https://go.dev/doc/effective-go#concurrency>
- [19] L. Anthony and Wails Contributors, "Wails - Build Desktop Applications Using Go," [Online]. Available: <https://wails.io/>
- [20] L. Anthony and Wails Contributors, "Wails v2 Documentation: Binding," [Online]. Available: <https://wails.io/docs/reference/runtime/intro>
- [21] R. Harris and Svelte Contributors, "Svelte - Cybernetically Enhanced Web Apps," [Online]. Available: <https://svelte.dev/>
- [22] R. Harris, "Rethinking reactivity," in *YGLF (You Gotta Love Frontend) Conference*, Vilnius, Lithuania, 2019.
- [23] S. Chacon and B. Straub, *Pro Git*, 2nd ed. Apress, 2014. [Online]. Available: <https://git-scm.com/book/en/v2>
- [24] GitHub, Inc., "GitHub: Where the World Builds Software," [Online]. Available: <https://github.com/>